THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Robert Hooke	In thin slices of cork, 1665.
2	**(a)** tissue	Tissue.
3	**(d)** biodiversity	Diversity of life.
4	**(a)** asexual reproduction	No gametes involved.
5	**(b)** biosphere	Sphere of life.
6	**(b)** prokaryotic cell	e.g. bacteria.
7	**(b)** sclerenchyma	Dead, with lignin.
8	**(d)** algae	Thallophyta.
9	**(a)** stamen	Anther and filament.
10	**(c)** fresh snow	High albedo.
11	**(d)** rough ER	Protein synthesis.
12	**(d)** epithelial tissue	Covering.
13	**(c)** Porifera	Pores on the body.
14	**(b)** fruit	Fruit.
15	**(a)** stratosphere	About 15 to 35 km.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Robert Hooke in 1665 (1); Box-like compartments in a cork slice (1).	1+1
17	Group of similar cells doing a specific function (1); Division of labour makes bodies efficient (1).	1+1
18	Variety of living organisms on the Earth (1); Classification makes the study of the large variety easy (1).	1+1
19	Production of new individuals of the same species (1); Ensures the continuity of species (1).	1+1
20	Geosphere: rocks and soil; hydrosphere: water (1); Atmosphere: gases; biosphere: life (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Raisin in water swells due to endosmosis (1); Raisin in strong sugar solution shrinks due to exosmosis (1); Water moves through the selectively permeable membrane (1).	1+1+1
22	Grow onion roots in water (1); Measure the length daily (1); Growth takes place near the tip due to apical meristem (1).	1+1+1
23	Cell type (prokaryotic/eukaryotic) (1); Body organisation (unicellular/multicellular) (1); Mode of nutrition and presence of cell wall (1).	1+1+1
24	Nucleus divides into two (1); Cytoplasm divides (1); Two daughter Amoebae form (1).	1+1+1
	OR: Evaporation from water bodies (hydrosphere) (1); Condensation and rain from the atmosphere (1); Rain affects soil (geosphere) and plants (biosphere) (1).	1+1+1
25	Shape suits the function (1); RBC: disc-shaped, carries oxygen (1); Nerve cell: long and branched, carries messages (1).	1+1+1
	OR: Cell body with nucleus (1); Dendrites receive impulses (1); Axon carries the impulse away (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Cell membrane (1); Nucleus with chromosomes (1); Cytoplasm with ER and Golgi (1); Mitochondria, lysosomes (1); Ribosomes and vacuoles (1).	1+1+1+1+1

	OR: Apical: tips, growth in length (1); Lateral: girth (1); Intercalary: base of leaves/internodes (1); Cells dividing, thin walls (1); Differentiate to form permanent tissues (1).	1+1+1+1+1
27	Monera (1); Protista (1); Fungi (1); Plantae (1); Animalia (1).	1+1+1+1+1
	OR: Fission (1); Budding (1); Fragmentation (1); Spore formation (1); Vegetative propagation (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** Schleiden and Schwann	Plants and animals are made of cells.
2	**(d)** striated (voluntary) muscles	Skeletal.
3	**(d)** gills	Aquatic.
4	**(c)** testes	Male organ.
5	**(c)** water (hydrological) cycle	Cycle of water.
6	**(d)** bacterium	No nuclear membrane.
7	**(b)** meristematic tissue	Dividing cells.
8	**(b)** taxonomy	Naming and grouping.
9	**(a)** binary fission	Splits into two.
10	**(a)** geosphere	Rocky part.
11	**(b)** lipid synthesis and detoxification	Fats and detox.
12	**(d)** xylem	Tracheids and vessels.
13	**(a)** bryophytes	Need water for fertilisation.
14	**(d)** carpel (pistil)	Stigma, style, ovary.
15	**(c)** ocean water or forests	Low albedo.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Chloroplast: photosynthesis (1); Chromoplast: colour; leucoplast: storage (1).	1+1
17	Xylem carries water and minerals (1); Phloem carries food (1).	1+1
18	Heterotrophic, cell wall of chitin (1); Saprophytes; e.g. yeast and mushrooms (1).	1+1
19	By spores from sporangia (1); Spores germinate on moist bread (1).	1+1
20	Cyclic movement of nutrients through living and non-living parts (1); Examples: carbon and nitrogen cycles (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Porifera: pores, canal system, sessile (1); Cnidaria: stinging cells, radial symmetry (1); Examples: Sycon; Hydra (1).	1+1+1
22	Insect: coloured, fragrant, nectar (1); Wind: small, light pollen, feathery stigma (1); Adaptations (1).	1+1+1
23	At the equator rays are nearly vertical (1); At the poles they are slanting (1); Spread over a larger area, less heating (1).	1+1+1
24	Store water and nutrients (1); Maintain turgidity (1); Store waste products (1).	1+1+1
	OR: Voluntary: controlled by will, e.g. biceps (1); Involuntary: not controlled, e.g. heart, intestine (1); Different tissues (1).	1+1+1
25	(a) Cnidaria; (b) Platyhelminthes (1); (c) Arthropoda (Insecta); (d) Amphibia (1); (e) Mammalia (1).	1+1+1
	OR: Barrier: condom (1); Chemical: pills (1); Surgical: vasectomy/tubectomy (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Four spheres (1); Energy from the Sun (1); Matter cycles (1); Interactions (1); Human influence (1).	1+1+1+1+1
	OR: Mitosis for growth and repair (1); Meiosis for gamete formation (1); Chromosome numbers (1); Daughter cells (1); Significance (1).	1+1+1+1+1
27	Striated (1); Smooth (1); Cardiac (1); Voluntary or involuntary (1); Locations (1).	1+1+1+1+1
	OR: Cotyledons (1); Roots (1); Leaf venation (1); Flower parts (1); Examples (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** new cells arise from pre-existing cells	All cells come from cells.
2	**(a)** squamous epithelium	Flat cells.
3	**(c)** Cnidaria (Coelenterata)	Stinging cells.
4	**(c)** fertilisation	Zygote formation.
5	**(c)** harmful ultraviolet radiation	Absorbs UV.
6	**(c)** mitochondrion	Releases energy.
7	**(d)** smooth (involuntary) muscles	Unstriated.
8	**(a)** Amphibia	Live on land and water.
9	**(a)** ovaries	Female organ.
10	**(d)** cycling of nutrients through living and non-living parts of the Earth	Matter recycles.
11	**(b)** DNA and proteins	Hereditary material.
12	**(c)** intercalary meristem	At the base of leaves/internodes.
13	**(c)** Carolus Linnaeus	Two-name system.
14	**(c)** budding	Buds from parent.
15	**(d)** hydrosphere	Water.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Shrinking of the cytoplasm away from the cell wall (1); In a hypertonic solution (1).	1+1
17	Protection (1); Absorption and secretion (1).	1+1
18	Platyhelminthes are flat, nematodes are cylindrical (1); Examples: tapeworm and Ascaris (1).	1+1
19	Many eggs and sperms are lost in the water (1); To increase the chance of fertilisation (1).	1+1
20	Provides gases like oxygen and carbon dioxide (1); Protects from UV rays and keeps the Earth warm (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Convert nitrogen into nitrates (1); Enrich the soil (1); Reduce the need for fertilisers (1).	1+1+1
22	Hollow a potato and put sugar solution in it (1); Place in water (1); Water level in the cavity rises by osmosis (1).	1+1+1
23	Leaf epidermis: stomata, cuticle (1); Root epidermis: root hairs absorb water (1); Protects underlying tissues (1).	1+1+1
24	They feed on dead and decaying matter (1); Break it down and recycle nutrients (1); Saprophytic nutrition (1).	1+1+1
	OR: A bud grows on the parent (1); The bud develops and detaches (1); Forms a new individual (1).	1+1+1
25	Gases trap heat, keeping the Earth warm (1); Burning fuels and deforestation add more CO2 (1); Enhanced effect causes global warming (1).	1+1+1
	OR: Hypotonic: lower solute outside, cell swells (1); Hypertonic: higher solute outside, cell shrinks (1); Isotonic: equal, no change (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Parents (1); Gametes (1); Variation (1); Speed (1); Examples (1).	1+1+1+1+1
	OR: Fixation by bacteria and lightning (1); Absorption of nitrates (1); Passing through the food chain (1); Decay by decomposers (1); Return to the air (1).	1+1+1+1+1
27	Cell wall (1); Chloroplasts and other plastids (1); Large central vacuole (1); Common organelles (1); Functions (1).	1+1+1+1+1

	OR: Parenchyma (1); Chlorenchyma and aerenchyma (1); Collenchyma (1); Sclerenchyma (1); Locations and functions (1).	1+1+1+1+1

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